

A Bio-Inspired Computational Model for the3D Motion Estimation

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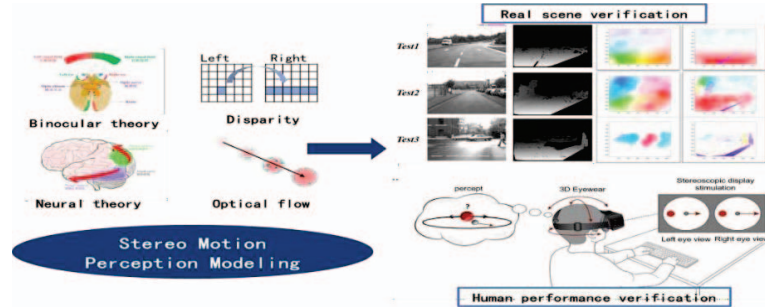


Figure 1: Illustrations of the stereo motion perception model creation and verification process. The model is built based on binocular theory and neural perception theory. Stereoscopic motion perception test is performed both on human and real scene to test the consistency and accuracy of the mode

ABSTRACT

Stereoscopic motion perception is pivotal for human-world interaction and has sparked advancements in artificial vision technologies. We introduce the Stereo Motion Perception Model (SMPM), a bio-inspired computational model designed to emulate the process of stereoscopic motion perception inherent in the human visual system. The SMPM extracts temporal, spatial, and shape features from stereoscopic video stimuli, enabling accurate motion perception. The performance of SMPM is tested and compared with human perception. The results indicates that the SMPM effectively estimates target motion on both simplistic and complex scene, high consistency is shown with human stereo motion perception. Future implications of the bio-inspired model is discussed.

Keywords: Stereoscopic motion perception, Stereo Motion Perception Model (SMPM), Directional asymmetry, Human visual system characteristics

1 INTRODUCTION

Motion perception is a fundamental aspect of human vision that allows us to navigate the world and interact with our surroundings. has led researchers to develop computational models that mimic the underlying mechanisms of the human visual system[1-3]. These models aim to improve the performance of artificial vision systems in applications such as robotics, autonomous vehicles, and virtual reality[4-7]. However, Among them, research on stereo motion perception is still very limited. The stereo perceptual ability arises from the integration of visual information from the two eyes, which provides depth cues that enable us to estimate the three-dimensional structure of our environment and track moving objects in space[8, 9].

We propose a stereo video motion perception model, Stereoscopic Motion Perception Model (SMPM), based on human eye visual characteristics for recognizing and evaluating the motion of moving

targets in stereo video. SMPM is designed to compute the motion of moving targets in space by extractingtemporal feature optical flow and spatial feature binocular disparity from stereo motion video stimuli separately and integrating perspective projection transformations(Figure 2). During the design of the model, the selective mechanism of human eye vision for motion and the macroscopic mechanism of human perceived motion are combined. We evaluated theperceptual effects of the model for detecting target motion in self- designed single-target synthetic scenes and natural scenes with stereo motion visual stimulus videos to test its adaptability to different scenes.

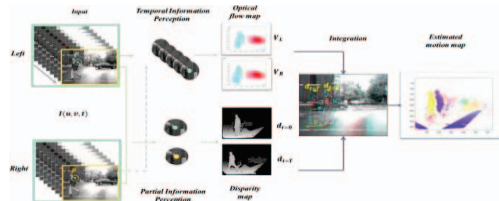


Figure 2: Illustrations of the 3D motion estimation process.

2 THEORY

The mechanism we use as reference in this paper is based on human vision system to obtain the location and direction of motion objects in three-dimensional space. In general, the mechanism of human visual perception can be explained by following three parts: human eye always receives external images to acquire the vector characteristics of motion changes in continuous time; human eye binocular channel receives images to acquire spatial features of binocular disparity; then human brain regions integrate image features to acquire motion perception.

Module 1: Temporal Information Perception

The goal of the temporal information perception module is to extract the representation of object motion over the temporal

dimension of 3D video in order to capture as much information as possible about the change of motion over time. We choose the most typical optical flow algorithm FF1MT to extract the temporal features of 3D video[10].

Module 2: Spatial Information Perception

Considering the high computational complexity of the FFV1MT model, we expect the stereo matching algorithm to have a lower computational cost. For this purpose, efficient large-scale Stereo algorithm (ELAS) is employed to calculate the binocular disparities of the video frames at every same moment.

Module 3: Spatial-Temporal Integration

The projective geometric transformation law shows that the correspondence between the image of a moving object in the real three-dimensional world $S(x, y, z)$ and the image projected on the retina $I(u, v)$ is a process of transforming three-dimensional coordinates into two-dimensional coordinates. Based on capturing the spatial-temporal features of motion in 3D video, we consider integrating the features through spatial transformation.

3 EXPERIMENT

In order to test the design effect of the model from different aspects, two experiments were designed in this paper. One is to elucidate the effect of the model on the human perception. The second one is to choose the stereo visual stimulus video of real scenes to explore the adaptability of the model.

Human stereoscopic perception: A total of 31 subjects participated in this experiment, with only one subject having stereo vision abnormalities, while the remaining 30 subjects had normal stereo vision and normal binocular correction. The goal of the participants is to intercept the grey ball by controlling the motion of the red ball, as shown in Figure 1.

Real scene motion estimation: We selected five different sets of stereoscopic motion visual stimulation videos to explore the estimation performance of SMPM.

4 RESULTS AND DISCUSSION

It indicates that in the stereoscopic motion visual stimulus perception task, the model and volunteers showed similar perceptual situations, and the volunteers' perceptual accuracy was higher than the model's perceptual accuracy. However, both the model and the volunteers showed perceptual asymmetry for different motion directions, and the gap in perceptual asymmetry was greater for the volunteers.

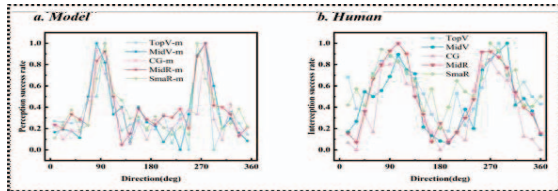


Figure 3a. Relationship between model perceived success rate and orientation 3b. Volunteers' perceived success in relation to orientation in our previous studies

Based on Figure 4, it can be seen that the motion visualization map of the model can represent the position and motion direction of the moving target relative to the observer. The model perceives well for single-target, empty visual scenes. When there are occlusions, complex textures, and light and darkness in the scene, they all interfere more with the perception of the model. In addition, the binocular disparity algorithm chosen in this paper has a hole in the lower right corner for the calculation of such

video disparity maps. This indicates that the physical model designed according to the projection variation is feasible, but it needs to be improved, and a more accurate and adaptive feature extraction model can be considered.

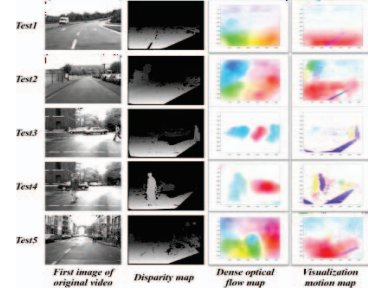


Figure 4 Application of the model to real scene stereo motion stimulation videos

5 CONCLUSION

In this paper, we propose a stereoscopic motion perception model based on the visual properties of the human eye. Experimental results on visual stimuli in real scenes show that our framework successfully estimates the motion of a moving target, providing a new perspective for designing models in a way that incorporates the characteristics of the human visual system. Moreover, there is much room for model improvement, and more efficient feature extraction models can be considered to compute accurate motion in combination with deep learning.

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